

Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

For a switched-affine plant following a prescribed fixed-period mode sequence, static cycle-to-cycle feedback can locally stabilize a reference periodic trajectory by shifting its $N - 1$ interior switching instants, but its raw switching-instant offsets may produce inadmissibly short or negative dwell durations. With strict nominal dwell margin, we multiply the complete raw vector by the largest conditioning factor in $[0, 1]$ that keeps every dwell duration admissible; the applied offsets preserve the requested direction, mode order, and cycle period. One scan of the N dwell changes computes the factor in $O(N)$ time, with no online solver. For the linearized raw closed loop, an offline polyhedral iteration finds the maximal subset of the Raw-Action Admissible Region that remains dwell-feasible at every future cycle; Schur stability then gives convergence inside this set. We derive the one-cycle linearization from exact augmented matrix-exponential propagation for mode-dependent affine dynamics. In a nine-interval DC-DC converter benchmark, the computed set has 7 facets and 10 vertices and equals the complete Raw-Action Admissible Region. An exact nonlinear 100-cycle simulation starts outside this set: the aggressive raw action produces a $-43.045 \mu\text{s}$ dwell, while conditioning enforces the assumed $3 \mu\text{s}$ bound and the trajectory enters the set at cycle 3. The invariant-set guarantee applies to the linearized raw closed loop. The exact nonlinear guarantee is local where conditioning is inactive, and the large-error result is simulation evidence.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, one-cycle map, DC-DC converter

1 Introduction

Power converters control voltage and current by moving through a sequence of switch configurations. In practice, the switching pattern must respect electrical limits, switching frequency, a minimum dwell duration for each configuration, and a short computation time [Karamanakos et al. \(2020\)](#). Keeping the individual configurations explicit gives a hybrid-affine or switched-affine model

[Albea-Sanchez et al. \(2021\)](#). Repeating the pattern produces a periodic trajectory rather than a constant state, and controlling such trajectories is an established problem for switched-affine systems [Egidio et al. \(2020\)](#); [Xu and Lazar \(2024\)](#). We consider regulation at a selected cycle anchor. At each cycle start, the controller steers the state toward that point on the reference periodic trajectory.

Predictive control tackles the problem by solving a constrained optimization problem online.

Recent work reduces the online burden through Laguerre parameterization, dedicated fixed-frequency solvers, and candidate pruning [Saeed et al. \(2022\)](#); [Yang et al. \(2022\)](#); [Augustine and Patil \(2023\)](#). Controllers also differ in what they may change. State-dependent switching laws choose the active subsystem [Egidio et al. \(2022\)](#), where switching-time control keeps an ordered sequence and moves its transition instants. We take this same approach, changing the transition instants and keeping the rest of the parameters the same. At each cycle start, the error is computed and the offsets applied during that cycle, with no online solver.

Patino et al. studied mode-dependent cyclic systems and how each switching instant affects the trajectory. They applied an online predictive controller to adjust the switching instants and track an optimal periodic trajectory [Patino et al. \(2010\)](#). Marcolino et al. considered a fixed-period cycle controlled by changing its interior switching instants. They used a discrete LQR and dwell-constrained predictive control to apply corrections based on the one-cycle model [Marcolino et al. \(2021\)](#). Their model has a common state matrix with switched actuator levels. Each interior switching instant separates two dwell durations, corresponding to the modes active before and after it. Delaying the instant lengthens the preceding dwell and shortens the following one, so the dwell limits are coupled. The predictive controller enforces these limits through online optimization, and its feasible set is characterized through polyhedral projection [Marcolino et al. \(2021\)](#). The standalone LQR comparison instead clips the offsets one at a time, which changes the requested direction and may shift the final scheduled instant [Marcolino et al. \(2021\)](#). Projecting a raw action directly onto the feasible set would require a quadratic program and would generally change its direction. Direction-preserving control allocation scales the complete command, but it also uses online optimization and does not address fixed-period dwell constraints [Nakamura-Zimmerer and Miller \(2026\)](#). The remaining problem is to find the largest feasible scale directly, preserving the timing-action direction, mode order, and cycle period without an online solver.

We adapt direction-preserving scaling to fixed-period switching schedules. Each dwell constraint

gives an upper bound on the scale, and the smallest bound is the largest feasible factor in $[0, 1]$. Checking all N bounds takes $O(N)$ time and requires no online solver. Scaling the complete raw offset vector preserves its direction, the mode order, and the cycle period. For the linearized raw closed loop, an offline polyhedral iteration finds the maximal invariant subset of the Raw-Action Admissible Region. Outside this set, the conditioner guarantees current-cycle dwell feasibility but not convergence; the nonlinear guarantee remains local. [Section 2](#) derives the cycle model, [section 3](#) develops the conditioner and invariant region, and [section 4](#) presents the converter study.

2 Fixed-period one-cycle model

The conditioner of [section 3](#) needs two things from the plant: how a switching-instant offset changes the next cycle-start error, and what dwell durations that offset produces. This section derives both, in four steps. We fix the timing coordinates, propagate one cycle exactly in an augmented state, expand that product to first order in the dwell changes, and rewrite the result in the independent switching-instant offsets.

2.1 Switched-affine cycle and timing coordinates

Consider

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n, \quad (1)$$

where each cycle follows the fixed mode sequence $(q_1, \dots, q_N)$ with nominal boundaries

$$0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T, \quad \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \quad (2)$$

The nominal schedule generates a reference periodic trajectory $\bar{\mathbf{x}}(t)$. Its cycle anchor $\mathbf{x}^* = \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}(T)$ is a fixed point of the nominal cycle, and we measure the cycle-start error against it as $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$.

Only the interior switching instants move:

$$t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \quad (3)$$

$$\delta\tau_{k,0} = \delta\tau_{k,N} = 0.$$

The vector $\delta\tau_k$ contains the $N - 1$ independent offsets. The corresponding N dwell changes satisfy

$$\delta d_k = D\delta\tau_k, \quad D_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j + 1, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

for $i = 1, \dots, N$ and $j = 1, \dots, N - 1$, so $\delta d_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}$. Moving one instant later lengthens the interval before it and shortens the interval after it by the same amount. Since $\mathbf{1}^\top D = 0$, the dwell changes sum to zero and every offset vector preserves the cycle period T . We write the exact one-cycle map as $\mathbf{x}_{k+1} = \mathcal{P}(\mathbf{x}_k, \delta\tau_k)$.

2.2 Exact augmented propagation and prediction

We suppress the cycle index while deriving the propagation equation. Define the augmented state and matrix

$$\chi(t) = \begin{bmatrix} \mathbf{x}(t) \\ 1 \end{bmatrix}, \quad \tilde{A}_{\sigma(t)} = \begin{bmatrix} A_{\sigma(t)} & \mathbf{b}_{\sigma(t)} \\ \mathbf{0}^\top & 0 \end{bmatrix}. \quad (5)$$

Then

$$\dot{\chi}(t) = \tilde{A}_{\sigma(t)}\chi(t). \quad (6)$$

For one cycle, let

$$0 = t_0 < t_1 < \dots < t_N = T, \quad (7)$$

$$\sigma(t) = q_i, \quad t_{i-1} \leq t < t_i.$$

and set

$$F_i = \tilde{A}_{q_i}, \quad \Delta t_i = t_i - t_{i-1}. \quad (8)$$

Propagation through interval i gives

$$\chi(t_i) = e^{F_i \Delta t_i} \chi(t_{i-1}). \quad (9)$$

Applying this relation from interval 1 to interval N yields the exact cycle propagation

$$\chi(t_N) = e^{F_N \Delta t_N} e^{F_{N-1} \Delta t_{N-1}} \dots e^{F_2 \Delta t_2} e^{F_1 \Delta t_1} \chi(t_0). \quad (10)$$

The rightmost factor acts first. The matrix order follows the order in which the trajectory visits the modes.

The nominal interval duration and its perturbation are

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1} = \bar{d}_i, \quad \Delta t_i = \Delta \bar{t}_i + \delta t_i. \quad (11)$$

The corresponding nominal cycle is

$$\bar{\chi}(t_N) = e^{F_N \Delta \bar{t}_N} e^{F_{N-1} \Delta \bar{t}_{N-1}} \dots e^{F_2 \Delta \bar{t}_2} e^{F_1 \Delta \bar{t}_1} \bar{\chi}(t_0). \quad (12)$$

Define the augmented error by

$$e(t) = \chi(t) - \bar{\chi}(t), \quad \chi(t_0) = \bar{\chi}(t_0) + e(t_0). \quad (13)$$

Subtracting the nominal propagation from the actual propagation gives

$$e(t_N) = e^{F_N(\Delta \bar{t}_N + \delta t_N)} \dots e^{F_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{\chi}(t_0) + e(t_0)) - e^{F_N \Delta \bar{t}_N} \dots e^{F_1 \Delta \bar{t}_1} \bar{\chi}(t_0). \quad (14)$$

2.3 First-order prediction in interval durations

Since F_i is constant on interval i ,

$$\frac{d}{ds} e^{F_i s} = F_i e^{F_i s}. \quad (15)$$

Therefore,

$$e^{F_i(\Delta \bar{t}_i + \delta t_i)} = e^{F_i \Delta \bar{t}_i} + F_i e^{F_i \Delta \bar{t}_i} \delta t_i + \mathcal{O}(\delta t_i^2). \quad (16)$$

Using

$$\phi_i = e^{F_i \Delta \bar{t}_i}, \quad (17)$$

the ordered product becomes, to first order,

$$\begin{aligned} & e^{F_N(\Delta \bar{t}_N + \delta t_N)} \dots e^{F_1(\Delta \bar{t}_1 + \delta t_1)} \\ & \doteq \tilde{\Phi} + \sum_{i=1}^N \phi_N \dots \phi_{i+1} F_i \\ & \quad \times \phi_i \dots \phi_1 \delta t_i, \\ & \tilde{\Phi} = \phi_N \phi_{N-1} \dots \phi_1. \end{aligned} \quad (18)$$

An empty product is the identity. In particular, the coefficient of δt_1 begins with $\phi_N \dots \phi_2 F_1 \phi_1$, whereas the coefficient of δt_N begins with $F_N \phi_N \dots \phi_1$. Matrices from different modes are not interchanged.

Terms $\delta t_i e(t_0)$ and $\delta t_i \delta t_j$ are second order. After cancelling the nominal propagation in (14), define

$$\begin{aligned}\delta \mathbf{t} &= [\delta t_1 \ \delta t_2 \ \cdots \ \delta t_N]^\top, \\ \gamma_i &= \phi_N \cdots \phi_{i+1} F_i \phi_i \cdots \phi_1 \bar{\chi}(t_0), \\ \Gamma &= [\gamma_1 \ \gamma_2 \ \cdots \ \gamma_N].\end{aligned}\quad (19)$$

Thus the columns of Γ follow the same order as the entries of $\delta \mathbf{t}$, and the augmented prediction equation is

$$\boxed{e(t_N) \doteq \tilde{\Phi} e(t_0) + \Gamma \delta \mathbf{t}}. \quad (20)$$

Here $\doteq$ denotes equality to first order about $e(t_0) = 0$ and $\delta \mathbf{t} = 0$.

To recover the physical-state equation, define

$$\Pi = [I_n \ \mathbf{0}], \quad \iota = \begin{bmatrix} I_n \\ \mathbf{0}^\top \end{bmatrix}. \quad (21)$$

At the cycle boundaries, $e(t_0) = \iota e_k$ and $\Pi e(t_N) = e_{k+1}$. Restoring the cycle index on the duration perturbation gives

$$e_{k+1} \doteq \Phi e_k + G_d \delta \mathbf{t}_k, \quad (22)$$

where

$$\begin{aligned}\Phi &= \Pi \tilde{\Phi} \iota = e^{A_{q_N} \Delta \bar{t}_N} \cdots e^{A_{q_1} \Delta \bar{t}_1}, \\ G_d &= \Pi \Gamma.\end{aligned}\quad (23)$$

Because every ϕ_i is block upper triangular,

$$\tilde{\Phi} = \phi_N \cdots \phi_1 = \begin{bmatrix} \Phi & \mathbf{h} \\ \mathbf{0}^\top & 1 \end{bmatrix}. \quad (24)$$

Thus $\tilde{\Phi}$ is the augmented propagation matrix, Φ is its physical $n \times n$ state-propagation block, and G_d maps the ordered interval-duration perturbations to the terminal physical-state perturbation.

2.4 Switching-instant offsets

The duration-perturbation vector in cycle k is $\delta \mathbf{t}_k = \delta \mathbf{d}_k$. Substituting $\delta \mathbf{t}_k = D \delta \boldsymbol{\tau}_k$ into (22) gives the model in the independent timing coordinates

$$\begin{aligned}e_{k+1} &= \Phi e_k + \Gamma_\tau \delta \boldsymbol{\tau}_k + \mathcal{O}(\| [e_k^\top \ \delta \boldsymbol{\tau}_k^\top] \|^2), \\ \Gamma_\tau &= G_d D.\end{aligned}\quad (25)$$

Column i of D increases dwell i and decreases dwell $i + 1$. Therefore $\Gamma_{\tau,i} = \Pi(\gamma_i - \gamma_{i+1})$. Since F_{i+1} commutes with $\phi_{i+1} = e^{F_{i+1} \bar{d}_{i+1}}$, factoring the adjacent terms yields

$$\begin{aligned}\Gamma_{\tau,i} &= e^{A_{q_N} \bar{d}_N} \cdots e^{A_{q_{i+1}} \bar{d}_{i+1}} \\ &\quad \times [(A_{q_i} - A_{q_{i+1}}) \bar{\mathbf{x}}(\bar{t}_i) + \mathbf{b}_{q_i} - \mathbf{b}_{q_{i+1}}]\end{aligned}\quad (26)$$

The bracket is the jump in the affine vector field at the switching state, from mode q_i to mode q_{i+1} . That is what displacing instant i does. It trades time between two adjacent modes without changing the state at that instant to first order, and the remaining nominal transitions carry the resulting jump to the end of the cycle. [Flieller et al. \(2006\)](#) also used switching-time sensitivities to compute hybrid limit cycles and test their sampled-time local stability. Equation (26) therefore has the switching-sensitivity structure reported in [Patino et al. \(2010\)](#), obtained here from dwell perturbations in independent fixed-period timing coordinates.

If every interval instead shares $A_{q_i} = A_c$ and $\mathbf{b}_{q_i} = B_c \mathbf{u}_{i-1}$, then

$$\Phi = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c (T - \bar{t}_i)} B_c (\mathbf{u}_{i-1} - \mathbf{u}_i), \quad (27)$$

which matches Marcolino's switched-actuator timing model, apart from the sign convention for the input jump [Marcolino et al. \(2021\)](#). When A_{q_i} depends on the mode, the jump also depends on the switching state, so this reduction does not apply in general.

3 Solver-free dwell-time conditioning

[Briat and Seuret \(2013\)](#) give stability tests for uncertain linear switched systems under minimum and mode-dependent dwell-time conditions. Here, [section 2](#) says what an offset does, not whether it can be applied. A static law knows nothing about the minimum dwell duration, and (4) ties each dwell to two adjacent offsets, so clipping the offsets one at a time would change the direction of the correction. We keep the direction and give up magnitude instead, then certify the closed loop that results.

Let a static feedback law produce the raw switching-instant offset

$$\delta\tau_k^{\text{raw}} = -K\mathbf{e}_k, \quad A_{\text{cl}} = \Phi - \Gamma_\tau K. \quad (28)$$

Let $d_{\min}$ be the required minimum dwell duration and assume that the nominal schedule keeps a strict margin above it,

$$\mathbf{m} = \bar{\mathbf{d}} - d_{\min}\mathbf{1} > \mathbf{0}. \quad (29)$$

Each m_i is how much dwell i can be shortened before it becomes inadmissible. The raw action is feasible for exactly those errors in the polyhedron

$$\mathcal{C} = \{\mathbf{e} : \bar{\mathbf{d}} - D\mathbf{K}\mathbf{e} \geq d_{\min}\mathbf{1}\}. \quad (30)$$

For $\mathbf{e}_k \in \mathcal{C}$ the raw action needs no modification. Outside $\mathcal{C}$ at least one dwell would fall below $d_{\min}$. Collect the requested dwell changes in $\mathbf{q}_k = D\delta\tau_k^{\text{raw}}$ and set

$$\beta_k = \min \left\{ 1, \min_{i: q_{k,i} < 0} \frac{m_i}{-q_{k,i}} \right\}, \quad (31)$$

$$\delta\tau_k^{\text{app}} = \beta_k \delta\tau_k^{\text{raw}}.$$

where a minimum over an empty set is $+\infty$. We call β_k the conditioning factor. It equals one whenever the raw action is already feasible, and otherwise the dwell that runs out of margin first decides it.

Proposition 1 (Feasibility and maximality). *Under (29), (31) returns the largest $\beta \in [0, 1]$ such that every applied dwell duration satisfies $\bar{\mathbf{d}} + D(\beta\delta\tau_k^{\text{raw}}) \geq d_{\min}\mathbf{1}$. The conditioned action preserves the cycle period and the direction of the raw action. Computing it takes $O(N)$ time and requires no online solver.*

Proof A dwell with $q_{k,i} \geq 0$ cannot be shortened by scaling. A dwell with $q_{k,i} < 0$ requires $\beta \leq m_i/(-q_{k,i})$. Intersecting these bounds with $[0, 1]$ gives exactly (31), which is therefore both feasible and maximal. The identity $\mathbf{1}^\top D = 0$ preserves T , and multiplying the full action by one nonnegative scalar leaves its direction unchanged. A single pass over the N dwell changes evaluates every bound. $\square$

The Raw-Action Admissible Region only guarantees the next schedule. To identify the errors for

which the raw law remains feasible at every future cycle, set $\mathcal{C}_0 = \mathcal{C}$ and iterate

$$\mathcal{C}_{j+1} = \mathcal{C}_j \cap \{\mathbf{e} : A_{\text{cl}}\mathbf{e} \in \mathcal{C}_j\}, \quad \mathcal{C}_\infty = \bigcap_{j=0}^{\infty} \mathcal{C}_j. \quad (32)$$

This preimage intersection is the standard maximal-admissible-set construction [Gilbert and Tan \(1991\)](#). For polyhedral $\mathcal{C}$, every iterate is polyhedral and redundant inequalities can be removed offline.

Proposition 2 (Invariant raw-action feasibility and convergence). *The set $\mathcal{C}_\infty$ in (32) is the maximal positively invariant subset of $\mathcal{C}$ under A_{cl} . If A_{cl} is Schur, then every linearized trajectory starting in $\mathcal{C}_\infty$ remains raw-action feasible, has $\beta_k = 1$ for all k , and converges to the cycle anchor.*

Proof Membership in $\mathcal{C}_\infty$ means $A_{\text{cl}}^j \mathbf{e}_0 \in \mathcal{C}$ for every $j \geq 0$. Thus the raw action satisfies the dwell constraints at every cycle, the conditioner returns one, and $\mathbf{e}_k = A_{\text{cl}}^k \mathbf{e}_0$ remains in $\mathcal{C}_\infty$. No larger subset of $\mathcal{C}$ has this property by the definition in (32). Since A_{cl} is Schur, $A_{\text{cl}}^k \mathbf{e}_0 \rightarrow \mathbf{0}$. $\square$

The set computation is offline and does not alter the controller. Outside $\mathcal{C}_\infty$, [proposition 1](#) still guarantees that the applied schedule satisfies the current cycle's dwell constraints. The conditioned linearized matrix is $(1 - \beta_k)\Phi + \beta_k A_{\text{cl}}$, but this set argument makes no convergence claim for trajectories while $\beta_k < 1$.

The nonlinear statement is also local. Suppose $\mathcal{P}$ is continuously differentiable while the mode order is preserved, and that A_{cl} is Schur. The strict nominal margin and $\delta\tau^{\text{raw}}(\mathbf{0}) = \mathbf{0}$ put a neighborhood of the anchor inside $\mathcal{C}$. Inside that neighborhood the conditioner never acts, $\beta = 1$, and the Jacobian of the exact closed-loop map is A_{cl} , so the exact map exponentially stabilizes the cycle anchor and the phase-locked reference periodic trajectory through it. This does not extend the polyhedral certificate to the exact nonlinear map over all of $\mathcal{C}_\infty$.

4 Converter case study

The case study asks three questions. Does the derived one-cycle model reproduce the exact plant?

What invariant raw-feasible region does the aggressive controller admit? And does conditioning deliver anything that an already-feasible controller cannot?

4.1 Three-cell converter benchmark and implementation

The hybrid-affine converter analysis of [Albea-Sanchez et al. \(2021\)](#) embeds PWM and sample-and-hold variables and uses exact two-subinterval matrix-exponential propagation in a Lyapunov analysis. The present case study uses a prescribed nine-interval mode sequence. We use the three-cell multilevel DC–DC converter studied by Patino et al. [Patino et al. \(2010\)](#), with state $\mathbf{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$,

$$A_{\mathbf{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix},$$

$$\mathbf{b}_{\mathbf{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (33)$$

Both $A_{\mathbf{u}}$ and $\mathbf{b}_{\mathbf{u}}$ depend on the switch state. The plant therefore does not satisfy the common-state-matrix assumption in (27). [Table 1](#) collects the converter, schedule, and feedback parameters used in this section.

The reported physical mode sequence is (0, 1, 3, 7, 2, 0, 4, 7, 4), the rounded cycle boundaries are (0, 66, 88, 110, 132, 154, 220, 242, 264, 286) μs , and the reported cycle anchor is $\mathbf{x}^* = (9.9247, 19.2928, 0.9823)^\top$. Because those values are rounded, exact propagation through them does not return to the anchor. We therefore adjust the dwell vector once by least squares, holding the published 286 μs period, the reported anchor, and the 22 μs nominal-design bound. The resulting schedule is

$$\bar{\mathbf{t}} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, 218.210, 240.380, 263.828, 286) \mu\text{s}. \quad (34)$$

The largest boundary correction is 2.111 μs , and the closure error in the infinity norm is 1.421e-14. The adjustment prepares the benchmark data. It is not a trajectory-design method.

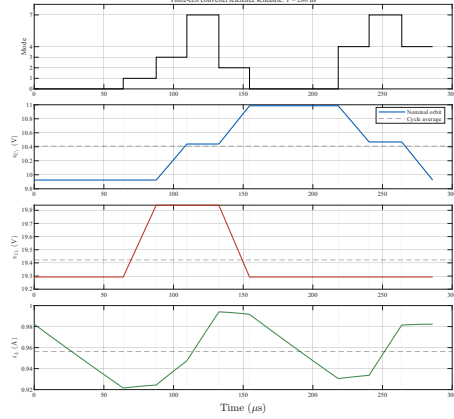


Figure 1 Reconciled reference schedule and exact periodic trajectory for the three-cell converter. Dashed horizontal lines show cycle averages. Dotted vertical lines mark the eight movable interior switching instants.

The cycle averages in [figure 1](#) are 10.410 56 V, 19.422 28 V, and 0.956 13 A. We design in the normalized coordinates $\mathbf{z} = S_x^{-1}\mathbf{e}$ and $\mathbf{v} = \delta\boldsymbol{\tau}/t_s$, which bring the three states and the eight offsets to comparable scale. The aggressive raw law is a standard discrete LQR with $Q_n = I_3$ and $R_n = 0.001I_8$. The conservative comparison law changes only R_n , to $0.1I_8$. Only the aggressive law is conditioned.

4.2 Linearization and invariant-region checks

We check the analytical Jacobians against central differences of the exact one-cycle map. The state steps are 10^{-4} times the diagonal entries of S_x , and the timing step is $10^{-3}t_s = 0.01 \mu\text{s}$. A second check applies simultaneous state and timing perturbations with normalized amplitude ϵ and evaluates

$$r(\epsilon) = \left\| S_x^{-1} [\mathcal{P}(\mathbf{x}^* + \Delta\mathbf{x}, \delta\boldsymbol{\tau}) - \mathbf{x}^* - \Phi\Delta\mathbf{x} - \Gamma_\tau\delta\boldsymbol{\tau}] \right\|_2. \quad (35)$$

The Jacobian errors and residual slope in [table 2](#) confirm the implemented first-order model. The relative errors sit at the level of numerical noise, and the residual decays with slope 2.000, the second-order decay a correct first-order model must show. The generated data include the full analytical and finite-difference matrices and the residual sweep.

Table 1 Converter, schedule, and feedback parameters. The applied bound of $3\text{ }\mu\text{s}$ is a simulation assumption, not the nominal benchmark-design bound.

Quantity	Symbol	Value
Source; capacitances	$E; C_1, C_2$	30 V; 40 μF
Inductance; resistance	$L; R$	10 mH; 10 Ω
Intervals; period	$N; T$	9; 286 μs
Nominal-design dwell bound	d_{bench}	22 μs
Applied-schedule dwell bound	d_{min}	3 μs
State; timing scale	$S_x; t_s$	$\text{diag}(10, 20, 1)$; 10 μs
Aggressive LQR weights	$Q_n; R_n$	I_3 ; $0.001I_8$
Conservative LQR weights	$Q_n; R_n$	I_3 ; $0.1I_8$

For the aggressive normalized closed loop, the Raw-Action Admissible Region has 7 facets and 10 vertices. One preimage intersection establishes that it is already positively invariant, so $\mathcal{C}_\infty = \mathcal{C}$. Its volume in physical cycle-start-error coordinates is 4.5410 $\text{V}^2 \text{A}$. Its coordinate ranges are $[-2.358, 2.098]$ V, $[-3.234, 1.911]$ V, and $[-0.496, 0.222]$ A. Every vertex satisfies the raw dwell constraints, and every vertex image lies strictly inside them with at least 4.347 μs margin. Together with $\rho(A_{\text{cl}}) = 0.568151 < 1$, these checks establish [proposition 2](#) for this linearized design.

To isolate the effect of action scaling, [figure 2](#) also shows the one-cycle feasible regions obtained by fixing $\beta = 0.5$ and $\beta = 0.2$. The dwell constraints depend on the product of β and the cycle-start error. Relative to $\beta = 1$, these factors multiply every radial extent by 2 and 5, and increase the three-dimensional volume by factors of 8 and 125. The corresponding volumes are 36.3282 and 567.6277 $\text{V}^2 \text{A}$. These outer regions describe feasibility for deliberately fixed factors; the actual conditioner still selects the largest feasible β_k from the current error.

4.3 Large-error simulation

We simulate the exact nonlinear map for 100 cycles from

$$\mathbf{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^\top. \quad (36)$$

From this state we run the conditioned aggressive law, the unconditioned conservative law, and the open-loop schedule. Exact matrix exponentials propagate every interval, so the simulation never

uses the linearized model. The initial normalized error is 0.983946, far outside the perturbation range used to check the local linearization.

Without conditioning, the aggressive law requests a minimum dwell of -43.045 μs , which would invert an interval and put two cycle boundaries out of order. Across the transient, the conditioner acts on 3 of the 100 cycles and drops to $\beta_{\text{min}} = 0.293978$, cutting the largest offset from 65.217 to 19.172 μs . The minimum applied dwell is 3.000 μs , exactly the assumed bound, which is what maximality predicts. The conservative law needs no conditioning and holds a minimum dwell of 15.502 μs , but it reaches the 0.01 error threshold only at cycle 20, against cycle 6 for the conditioned aggressive law.

The large-error initial condition lies outside $\mathcal{C}_\infty$, as its infeasible raw schedule implies. The conditioner is active for the first 3 cycles, and the exact nonlinear trajectory first enters the linearized invariant region at cycle 3. This entry is simulation evidence, not a claim that the nonlinear map must reach or preserve $\mathcal{C}_\infty$. [Figure 2](#) shows the region and the first ten conditioned cycle-start errors. Under the fixed nominal schedule, the open-loop cycle map has spectral radius $\rho(\Phi) = 0.999850 < 1$, so its cycle-start error also converges, but it remains above 0.01 until cycle 22009.

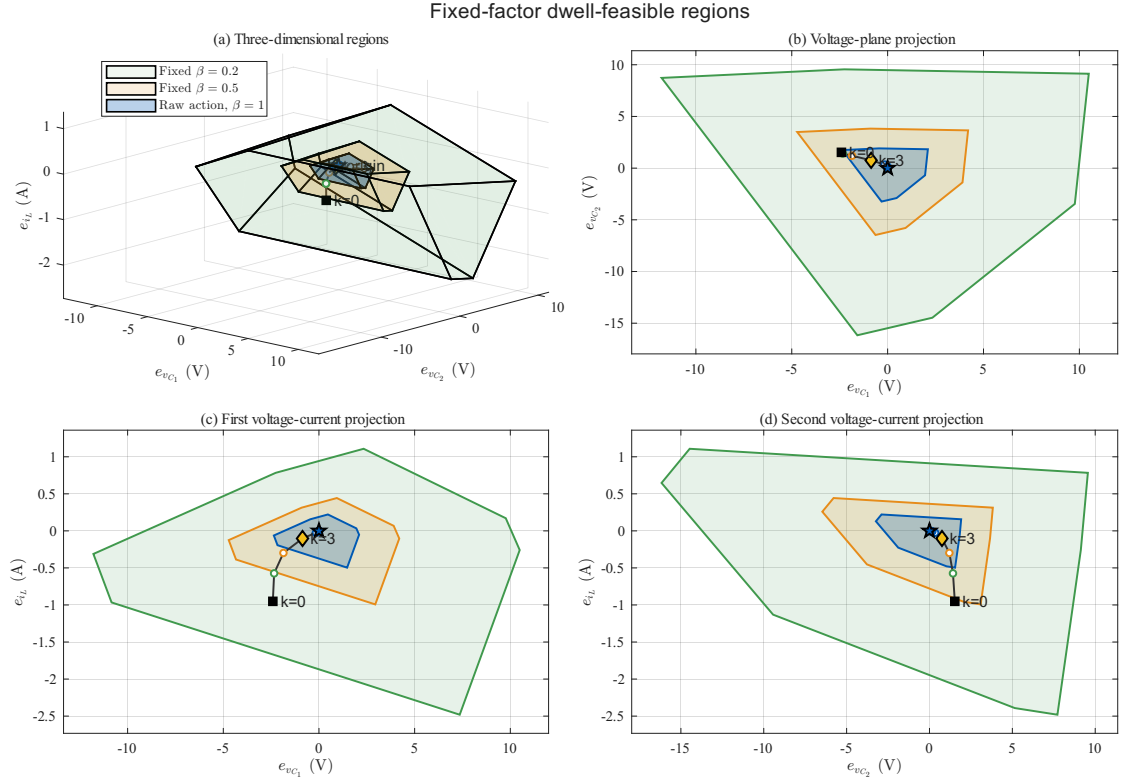


Figure 2 Fixed-factor dwell-feasible regions in physical cycle-start-error coordinates. (a) Three-dimensional overlay. (b)–(d) Orthogonal projections onto the coordinate planes. The blue, orange, and green regions use $\beta = 1$, $\beta = 0.5$, and $\beta = 0.2$, respectively. The blue region equals the invariant Raw-Action Admissible Region. Marker colors identify the smallest region containing each trajectory sample: green at $\beta = 0.2$, orange at $\beta = 0.5$, and blue for the raw action. The exact nonlinear conditioned trajectory first enters the raw-action region at cycle 3. Only membership in the full three-dimensional polyhedron certifies feasibility; the trajectory is simulation evidence.

Table 2 Linearization checks, invariant raw-action region, and exact nonlinear simulation results. Settling denotes the first cycle for which $\|S_x^{-1}e_k\|_2 < 0.01$.

Quantity	Value
Relative errors: Φ ; Γ_τ	8.518e-12; 8.344e-11
Linearization residual slope	2.000
Spectral radii: open; conservative; aggressive	0.999850; 0.941468; 0.568151
Invariant region: facets; vertices; volume ($V^2 A$)	7; 10; 4.5410
Initial state in region; first entry cycle	no; 3
Minimum conditioning factor; active cycles	0.293978; 3
Maximum raw; applied offset (μs)	65.217; 19.172
Minimum raw; applied dwell (μs)	-43.045; 3.000
Minimum conservative dwell (μs)	15.502
Settling cycle: conditioned; conservative	6; 20
Final errors: conditioned; conservative; open	1.746e-15; 3.113e-06; 0.298010

5 Discussion

The conditioner keeps the largest feasible scalar multiple of a static timing action and changes neither its direction nor the cycle boundaries. That is also its limit. It cannot look ahead, and it cannot choose a different direction when the requested one is nearly blocked, so it does not replace constrained optimization for controllers that must do either. The invariant region is also computed offline and adds no online work.

The cycle model joins two earlier formulations. Patino et al. derived the mode-dependent switching sensitivity [Patino et al. \(2010\)](#), and Marcolino et al. formulated fixed-period timing coordinates, dwell inequalities, and LQR for switched actuators [Marcolino et al. \(2021\)](#). Our derivation connects the two and checks the result numerically. Generic direction-preserving command scaling is established in control allocation [Nakamura-Zimmerer](#)

[and Miller \(2026\)](#). Our control contribution is its closed-form specialization to the coupled dwell inequalities, together with the invariant raw-action feasibility certificate; neither the generic scaling idea nor the LQR design is new here. The certificate is deliberately narrower than arbitrary conditioned stability: outside $\mathcal{C}_\infty$, feasibility is guaranteed for the applied schedule but convergence is supported only by simulation.

The 3 μ s minimum dwell is a simulation assumption rather than a hardware figure. The study leaves out timer quantization, delays, dead time, implementation margin, state estimation, model uncertainty, and experiments, and it fixes the mode order, phase, and period. The exact nonlinear guarantee holds only in a neighborhood where conditioning never acts, so the large-error simulation is evidence for this benchmark and not a global nonlinear result.

6 Conclusion

The conditioner enforces every minimum-dwell constraint in $O(N)$ time and keeps both the direction of the raw action and the cycle period. Offline preimage iteration identifies the maximal part of the Raw-Action Admissible Region that stays feasible under the linearized raw closed loop; for the converter design, the complete region is invariant and Schur stability gives convergence inside it. The augmented derivation covers mode-dependent switched-affine dynamics and reduces to the switched-actuator model when the state matrix is common, and finite differences confirm its Jacobians. In the exact nonlinear benchmark simulation, conditioning turns an infeasible aggressive action into an admissible one and the trajectory enters the computed region after three cycles. That entry and the behavior outside the region are simulation evidence. The nonlinear guarantee remains local, and hardware tests are still needed to account for timer quantization, delays, and implementation margin.

Declarations

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Data availability. TODO: state where the generated simulation data are available, or that all data supporting the findings are contained in the article. Required for original research.

Code availability. TODO: state where the simulation and figure-generation scripts are available, or that this item is not applicable.

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